AP CALCULUS AB

Unit 1 • Lesson 1.11 practice • Answer key

Lesson 1.11 • Defining Continuity at a Point

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Let $f(x) = x^2$ for $x \neq 2$ and $f(2) = 1$. Which condition for continuity at $x = 2$ fails?

(A) $f(2)$ is defined.
(B) $\lim_{x \to 2} f(x)$ exists.
(C) $\lim_{x \to 2} f(x) = f(2)$.
(D) None; f is continuous at 2.

Solution

Answer: (C). The limit is 4 but $f(2) = 1$.

Question 2

NO CALCULATOR

Let $f(x) = \begin{cases} 2x+1, & x < 1 \\ 3, & x = 1 \\ x^2+2, & x > 1. \end{cases}$ Is f continuous at $x = 1$?

(A) Yes
(B) No, $f(1)$ is undefined.
(C) No, the limit does not exist.
(D) No, the limit is not equal to $f(1)$.

Solution

Answer: (A). Left limit 3, right limit 3, and $f(1) = 3$.

Question 3

NO CALCULATOR

Why is $f(x) = \dfrac{x^2 - 9}{x - 3}$ not continuous at $x = 3$?

(A) The limit does not exist.
(B) $f(3)$ is undefined.
(C) The limit is infinite.
(D) It is continuous at $x = 3$.

Solution

Answer: (B). The denominator is 0 at $x = 3$, so $f(3)$ is undefined (the limit is 6).

Question 4

NO CALCULATOR

Which function is continuous at $x = 0$?

(A) $\dfrac{1}{x}$

(B) $\dfrac{|x|}{x}$

(C) $\dfrac{\sin x}{x}$

(D) $|x|$

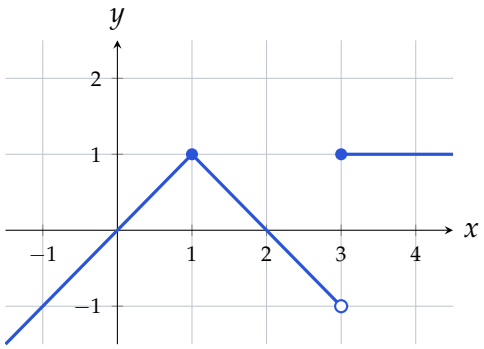
Solution

Answer: (D). $|x|$ is defined at 0 and its limit there is $0 = |0|$. The others are undefined at 0.

Question 5 (graph)

NO CALCULATOR

The graph of f is shown. Which statement is true?



(A) f is continuous at $x = 3$ but not at $x = 1$.

(B) f is continuous at $x = 1$ but not at $x = 3$.

(C) f is continuous at both $x = 1$ and $x = 3$.

(D) f is continuous at neither $x = 1$ nor $x = 3$.

Solution

Answer: (B). At $x = 1$ both pieces meet at $(1, 1)$: the limit is $1 = f(1)$. At $x = 3$ the left-hand limit is -1 and the right-hand limit is 1, so the limit does not exist.

Question 6 (Free response)

NO CALCULATOR

Let $f(x) = \begin{cases} x^2 - 1, & x < 2 \\ 3, & x = 2 \\ 5 - x, & x > 2. \end{cases}$

(a) Find $\lim_{x \to 2} f(x)$, showing both one-sided limits.

(b) Is f continuous at $x = 2$? Justify using the definition of continuity.

(c) Is f continuous at $x = 0$? Explain.

Solution

(a) $\lim_{x \to 2^-} f(x) = 4 - 1 = 3$ and $\lim_{x \to 2^+} f(x) = 5 - 2 = 3$, so $\lim_{x \to 2} f(x) = 3$.
(b) Yes: $f(2) = 3$ is defined, $\lim_{x \to 2} f(x) = 3$ exists, and $\lim_{x \to 2} f(x) = f(2)$.
(c) Yes. Near $x = 0$, $f(x) = x^2 - 1$, a polynomial, which is continuous everywhere.